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Dear Team,

Thank you for participating in the third edition of QualTech® Prize - Education.

This is a verified copy of the Project Report submitted by you and your team, applying the structured problem solving methodology from Qimpro's Problem Solving Pro digital course. It reflects the effort, rigour, and thinking each of you brought to your project and we hope you take pride in what you have accomplished together.

You may keep this report as a personal record of your achievement. It is also a powerful addition to your portfolio – tangible, verified evidence of hands-on experience in quality and problem solving that you can present to prospective employers and recruiters with confidence. With this experience, problem solving is no longer a buzzword on your résumé. It is something you have done, documented, and delivered. That is what builds a professional narrative. That is what the Junior Qualitist® credential stands for.

We hope this experience has strengthened your confidence, sharpened your skills, and that your Junior Qualitist® credential serves as a stepping stone toward many more accomplishments ahead.

From all of us at Qimpro® Foundation, we wish you the very best for your future endeavours.

We look forward to seeing all that you go on to achieve.

Warm regards,

JIGISHA LULLA
CEO

FATEMA KERA
Manager





EDUCATION

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5th Edition 2026 SAKCON CAHO

PROJECT REPORT

AJ Institute of Hospital Management
Mangalore

Project Title:
Improving Patient Care Attender Response Through a Digital Task Management System

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Project Title: Improving Patient Care Attender Response Through a Digital Task Management System

Introduction:

Personal Care Assistants (PCAs) deliver essential non-clinical support—such as hygiene assistance, sample transport, and bed preparation—directly impacting patient dignity, nursing efficiency, and overall workflow. To address systemic delays, this quality improvement study analyzed 5,877 task transactions across 39 hospital wards and departments over a 10-day period. By evaluating the root causes of delayed and cancelled requests, the study sought to replace the reactive workflow with a standardized, technology-driven allocation process, ensuring faster, more reliable task completion and optimized patient care.

Problem Definition:

The hospital experienced delays and cancellations in PCA tasks due to phone-based allocation, lack of real-time availability visibility, inconsistent task sequencing and inadequate cancellation-reason documentation. At baseline, 30.3% of timed tasks breached pickup TAT, while 86.2% of cancellations lacked a genuine, structured reason, creating avoidable delays and operational inefficiency.

Problem Diagnosis:

The problem diagnosis was conducted using the hospital's PCA transaction data, current-state process mapping and multiple quality improvement tools. A total of 5,877 PCA tasks recorded between 1, and 10 June 2026 were analysed across 27 task categories and 39 origin wards or units. The analysis focused primarily on task pickup turnaround time and cancellation-reason quality because these were the areas supported by verifiable, timestamped data.

The baseline analysis showed that 1,748 of 5,761 timed tasks, or 30.3%, breached their pickup target. The average Booking-to-Pickup time (T2) was 14.6 minutes, while Booking-to-Closure (T1) averaged 24.1 minutes. Emergency responsiveness was particularly concerning only 50.8% of Emergency tasks were picked up within the five-minute target, compared with 70.6% compliance for Normal-priority tasks within 15 minutes.

The current-state process map identified four major structural gaps. First, PCA availability was not verified before dispatch, requiring nurses to call and wait for a response. Second, when a PCA was busy, this information was not formally recorded, resulting in repeated calls and hidden TAT accumulation. Third, there was no standardized rule governing which task should be handled next. Finally, completion was not digitally confirmed, leaving nurses without real-time visibility of PCA status and increasing the likelihood of follow-up calls and delays.

Cancellation analysis revealed another important problem. Of 123 cancelled tasks, only 17 had genuine, structured reasons. 106 cancellations, or 86.2%, were not diagnostically useful. These included 43 entries recorded simply as "Other," 39 transporter names and 24 ward names instead of actual reasons. Therefore, the hospital was not only experiencing cancellations but was also unable to reliably determine why they occurred.

Pareto analysis identified the highest-impact task categories. Bed Pan (PCA) contributed 19,744 cumulative delay minutes, while Carbolization (PCA) contributed 18,368 minutes. Together they accounted for 45.8% of the analysed delay minutes, making them the priority categories for intervention. Lab Sample Collection and Bed Pan (MNO) were also major contributors, with the first four categories together accounting for 70.1% of cumulative delay.

Ward-level variation further demonstrated inconsistency in the process. Ward 5th B had a mean pickup delay of 29.7 minutes, while 3rd C had 22.5 minutes. In comparison, 7th C and 2nd Lab recorded much lower mean delays of 7.6 and 8.8 minutes respectively. This variation suggested that local escalation practices and hand-off discipline were important contributors rather than simply differences in task type.

The fishbone analysis categorized root causes under Men, Method, Measurement and Machine/Technology, confirming that the problem was primarily systemic rather than attributable to individual PCA performance.

Problem Remedy:

The problem remedy was designed directly from the identified root causes. Four interventions were developed and are being piloted over 12 weeks in the two priority wards, 5th B and 3rd C, focusing initially on Bed Pan–PCA and Carbolization–PCA tasks.

The first intervention is real-time PCA availability and auto-dispatch. Instead of relying on verbal confirmation, PCA status will be visible as Available, On-task or En-route, allowing nurses to identify available personnel before assigning a task. This aims to eliminate repeated calls and reduce allocation latency.

The second intervention is structured cancellation-reason capture. The existing free-text or blank cancellation field will be replaced with a mandatory dropdown containing standardized reasons such as patient unwillingness, postponement, duplicate booking, PCA unavailable for more than 15 minutes and alternate arrangements. This will improve the quality of data available for future root-cause analysis.

The third intervention introduces standard task sequencing and escalation rules. Emergency tasks will take priority at appropriate break points, while a PCA remaining busy for more than five minutes will trigger escalation to a second-call PCA.

The fourth intervention is a live TAT dashboard at nurse stations displaying daily mean T2 against target and current PCA assignment status.

Together, these interventions aim to improve response speed, standardize workload allocation, reduce avoidable cancellations and provide real-time operational visibility.

Locking the Improvements:

The strategy transitions Patient Care Assistant (PCA) allocation from a reactive, telephone-based system to a standardized, visible, and data-driven process. Baseline findings reveal significant operational variation between wards, yet top-performing units demonstrate that major turnaround time (TAT) improvements are entirely achievable within the current staffing model. The immediate objective is to reduce overall TAT breaches from 30.3% to below 10%, lower mean T2 response time from 14.6 to eight minutes or less, and increase both Emergency and Normal-task compliance to at least 90%. Additionally, cancellation-reason documentation quality is targeted to rise from 13.8% to at least 85%. To strengthen execution, real-time allocation will monitor supervisor booking times with a target under 15 seconds, while dynamic status visibility will promote balanced workload distribution across shifts. Once reliable baselines are established, secondary indicators like delayed sample collection, missed-care events, and gender-matching compliance will integrate into the operational dashboard. Beyond clinical efficiency, economic impact remains a core focus. The study estimates a baseline Cost of Poor Quality (COPQ) of ₹1.59–1.75 lakh per month due to avoidable delays. The pilot projects a recovery of 480–550 PCA hours monthly, translating to indicative cost savings of ₹1.15–1.30 lakh per month, subject to validation against actual HR and Finance wage data. Ultimately, performance will be evaluated through a holistic mix of operational, care-related, and financial metrics rather than TAT alone.

Quality Tools Used:

The Fishbone Cause-and-Effect Diagram was the primary quality tool used for problem diagnosis. It was developed around four major categories: Men, Method, Measurement, and Machine/Technology. The purpose was to systematically identify the underlying causes contributing to delayed PCA task pickup and unrecorded cancellations rather than attributing the problem to individual staff members.

The Men category highlighted inconsistent hand-offs and escalation practices. The Method category identified the absence of standardized task sequencing and escalation rules. Measurement highlighted inadequate real-time monitoring, lack of meaningful cancellation-reason captures and limited visibility of PCA status. Machine/Technology reflected the absence of real-time availability checking, digital closure confirmation and live dashboards. Other quality tools supported the diagnosis. A process map was used to understand the 10-step current-state workflow and identify points without verification or standardization. Check-sheet-style categorization converted the raw 5,877 task records and 123 cancellation records into usable categories. A histogram identified high-impact task categories, while a line/trend chart demonstrated daily variation in pickup delay. Two Pareto analyses prioritized the major sources of delay and the poor quality of cancellation data.

The fishbone therefore acted as the bridge between diagnosis and intervention: each major branch was translated into a corresponding corrective action.

Results:

At the current stage of the project, the 12-week intervention pilot has not yet completed, and therefore final post-improvement quantitative results are not available. The study deliberately avoids presenting projected targets as achieved outcomes.

The baseline results established a significant improvement opportunity. Among 5,761 timed tasks, 1,748 tasks (30.3%) breached their pickup TAT. Emergency-task compliance within the five-minute target was 50.8%, while Normal-task compliance within 15 minutes was 70.6%. Mean Booking-to-Pickup time across tasks was 14.6 minutes.

Cancellation analysis showed that only 13.8% of cancelled tasks had a genuine, structured reason, while 86.2% were associated with generic or incorrect entries. This demonstrated a major information-quality gap and became one of the key targets of the intervention.

The two highest-impact categories were Bed Pan–PCA and Carbolization–PCA, which together accounted for 45.8% of cumulative delay minutes. Ward 5th B recorded the highest mean pickup delay at 29.7 minutes, followed by 3rd C at 22.5 minutes.

Early implementation feedback from 5th B has indicated a nurse-reported reduction in repeat calls to the PCA desk, but this remains qualitative and has not been converted into a quantitative improvement claim.

The results should compare the same baseline measures with post-pilot performance, including TAT, cancellation-reason quality, workload distribution, allocation latency, care-related indicators and economic impact.

Lessons Learned:

The strategy transitions Patient Care Assistant (PCA) allocation from a reactive, telephone-based system to a standardized, visible, data-driven process. Baseline findings reveal significant ward variations, yet top performers prove that major turnaround time (TAT) gains are achievable within existing staffing. Key operational targets include reducing overall TAT breaches from 30.3% to below 10%, cutting mean T2 response time from 14.6 to eight minutes or less, and boosting Emergency and Normal-task compliance to at least 90%. Cancellation-reason documentation quality must also rise from 13.8% to 85%.

Execution will be strengthened by capping supervisor real-time booking times at under 15 seconds and leveraging dynamic status visibility to balance shift workloads. Future dashboard updates will incorporate tracking for delayed sample collections, missed-care events, and gender-matching compliance once baseline data stabilizes.

Financially, avoidable delays represent a baseline Cost of Poor Quality (COPQ) of ₹1.59–1.75 lakh per month. The pilot aims to recover 480–550 PCA hours monthly, yielding an estimated saving of ₹1.15–1.30 lakh per month, pending HR and Finance validation. Ultimately, success will be measured through a holistic mix of operational, care-related, and economic metrics rather than TAT alone.

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Improving Patient Care Attender Response Through a Digital Task Management System

12 September 2026

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